

TRIASHA SARKAR

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[linkedin.com/in/triasha-sarkar](https://www.linkedin.com/in/triasha-sarkar) · github.com/triasha72 · triasha72.github.io/Portfolio

MS in Aerospace Engineering (Georgia Tech) with a research background in statistical modeling, machine learning, and uncertainty quantification. Experience designing and auditing experiment campaigns of 15,000+ cases, controlling for data leakage, and diagnosing silent failure in data-generating pipelines. Prior industry experience validating safety-critical engineering models at Rolls-Royce.

EDUCATION

MS, Aerospace Engineering | Georgia Institute of Technology, Atlanta, GA

Aug 2026 (08/26)

- Graduate Research Assistant, Aerospace Systems Design Laboratory (ASDL), 2025–2026.
- Coursework: ISYE 6739 Statistical Methods (A); ISYE 6414 Regression Analysis.

MSc, Aerospace Vehicle Design | Cranfield University, United Kingdom

Feb 2023

First Class Honours · Thesis: *geometry optimization under FAA FAR 91.817 constraints in a data-scarce, simulation-expensive design space (SENECA consortium: Rolls-Royce, DLR, ONERA).*

B.Tech, Aerospace Engineering | SRM Institute of Science and Technology, India

Jun 2021

First Class Honours.

TECHNICAL SKILLS

Programming: Python (primary), MATLAB (proficient), C++ (working knowledge)

Data & ML: pandas, NumPy, SciPy, scikit-learn, PyTorch, pyDOE2, NetworkX, Jupyter, Git/GitHub

Statistics & Modeling: Regression and statistical inference; Gaussian process regression; design of experiments; uncertainty quantification and calibration; time-series anomaly detection; clustering; neural networks and CNNs; surrogate and reduced-order modeling; proper orthogonal decomposition; Operator Inference; multifidelity and Monte Carlo methods

RESEARCH & QUANTITATIVE PROJECTS

Silent Failure Detection in Large-Scale Simulation Campaigns

May – Jul 2026

Georgia Tech ASDL (AE8900) · Advised by Dr. Adam Cox

- Audited a **15,120-case design-of-experiments campaign** and found that 6,804 runs (45%) terminated with no output and no error message — failures that passed every standard validity check and would have silently biased every model built downstream.
- Validated labels against the full burn archive before modeling: every case wrote all ten output metrics or none, with zero exceptions, establishing the failure label as deterministic rather than noisy.
- Built a leakage-free classifier predicting failure from the **five pre-run geometry inputs alone**, with no post-run data admitted as a feature. A naive random split scored **99.8%** by memorising geometries duplicated nine times over; collapsing to the 1,680 unique combinations and testing only on geometry never seen in training gave the honest figure — **96.8% accuracy at 0.986 ROC AUC**, against a 55.0% majority-class baseline. The more impressive result was the wrong one.
- Replicated the result on an independent 1,050-case Latin hypercube study under a different sampling scheme (AUC 1.0) and identified a second failure mode at small absolute outer diameter.
- Traced the root cause to an uncapped pressure-balance loop in the solver source — a timeout counter incremented but never checked — and wrote and tested a patch converting silent hangs into logged convergence errors with full diagnostics.
- Stage 2 — profile classification over the 8,316 clean thrust-time histories — scoped, not yet implemented.

Cross-Sectional Equity Signal: Walk-Forward Backtest

Jul 2026

Independent · github.com/triasha72/Equity-Backtest

- Built a monthly, dollar-neutral cross-sectional equity backtest over **11,304 stock-months** of US large caps (2014–2024): expanding-window validation rather than k-fold, features standardised within each cross-section, and no post-formation data admitted as a model input. A shuffled-target test in the suite asserts the pipeline earns nothing when labels are permuted.
- Found that the momentum signal does not survive transaction costs — and was not significant before them: **net Sharpe 0.25 (t = 0.81)**, rising only to t = 1.07 with costs set to zero, against an equal-weight universe Sharpe of 1.00.
- Logged and published **all six specifications tested** rather than the best one. The strongest (t = 1.58) is reported as noise against a t = 3.0 hurdle stated before the result (Harvey, Liu & Zhu, 2016), and traced to portfolio diversification rather than predictive power.

Surrogate Model Comparison: Gaussian Process, Response Surface, RBF

2025 – Present

Independent · github.com/triasha72/Surrogate-model-learning

- Compared Gaussian process, response-surface, and radial-basis-function regressors across benchmark functions.
- Characterized how model assumptions drive error, uncertainty behavior, and failure modes in nonlinear regimes — where each estimator breaks down and why.

Cultural and Attitudinal Predictors of Life Expectancy

Summer 2026

Georgia Tech, ISYE 6414 Regression Analysis · Team of four

- Cross-national regression study over 79,143 World Values Survey responses across 53 countries, combining WHO and World Bank structural indicators with attitudinal measures; final model adjusted R^2 0.8097, test RMSE 2.245 years.
- Contributed the literature synthesis and cross-model integration — reconciling observation counts, country definitions, and variable specifications across three independently fitted models — and authored the abstract, introduction, and conclusions.
- With $n \approx 79,000$, conventional p-values flag nearly everything; variables were instead selected by **consensus across 100 subsamples under six criteria** (stepwise AIC/BIC, LASSO, elastic net). Documented pseudo-replication and ecological-fallacy limits constraining what the attitudinal coefficients can support.

Network Disruption Modeling: Atlanta Mobility Digital Twin

2026 – Present

Independent · github.com/triasha72/atlanta-mobility-resilience-digital-twin

- Graph model of Atlanta's road network (OSMnx, NetworkX, pandas) simulating disruption scenarios across origin–destination pairs and measuring travel-time and accessibility outcomes under stress.

NURBS-BEM Solver with Learned Surrogates

Oct 2025 – Present

Independent · github.com/triasha72/NURBS_BEM_EMSolver

- Implemented a mesh-free boundary-element solver from mathematical formulation through code; generated parametric multifidelity datasets for surrogate modeling, POD feasibility analysis, and Operator Inference.

Demand and Price-Sensitivity Modeling under Uncertainty (Project EAGLE)

Sep 2025 – Apr 2026

Vehicle Grand Challenge: RIC Enterprise / Georgia Tech ASDL

- Demand modeling and fare-sensitivity analysis for a 750+ passenger transport concept, including fleet-sizing across uncertainty scenarios.

Techno-Economic Modeling under Parametric Uncertainty (Project GREEN TEA)

Sep 2025 – Apr 2026

Georgia Tech ASDL / U.S. Endowment for Forestry

- Techno-economic and well-to-wake life-cycle assessment of alternative fuel pathways under parametric uncertainty; benchmarked model outputs against GREET, CORSIA, and 40BSAF-GREET reference frameworks.
- The resulting model is **in active use by the sponsor**, the U.S. Endowment for Forestry.

INDUSTRY EXPERIENCE

Systems Engineer | Alten India Pvt Ltd, embedded at Rolls-Royce, Bangalore

Oct 2023 – Apr 2025

- Supported airworthiness certification of the **Trent XWB-84 EP** engine; validated engineering models against certification standards under regulatory scrutiny.
- Worked directly with the consequences of model failure in deployed systems — the experience motivating subsequent research on silent failure detection.

LEADERSHIP

Graduate Chair, Women of Aeronautics and Astronautics (WoAA), Georgia Tech Chapter

2025 – 2026